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Produce While Explore Strategy: Phased Approach to Accelerate De-Risking and Monetization of Marginal Resources in West Abu Dhabi

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Abstract

In frontierbasin exploration, discoveries are often associated with major risks. The petroleum system elements are often not fully understood, or commercial aspects such as hydrocarbon recovery may not be straightforward. Therefore, development plans are normally not considered until sizable commercial discoveries are proven and de-risking through appraisal activities has been undertaken to mature the resources post-discovery. Consequently, the exploration-development strategy tends to be associated with lengthy and very expensive funding programs taking more than 10 years to witness the first hydrocarbon. For mature basins with marginal prospective resources, however, adopting a similar approach can significantly reduce the attractiveness of these opportunities. The West Abu Dhabi, which is regionally part of Rub' Al Khali mature basin, contains a stack of plays with either gas, condensate or oil resources that are trapped structurally in small four-way dip closures or stratigraphically in heterogeneous carbonate-evaporite deposits. The small size and geological nature of the resources (leads/prospects) require a different strategic mindset, where the Produce While Explore Strategy needs to be of a different form.

The strategy adopted for the West Abu Dhabi area hinges on the Explore-Produce concept to facilitate quick cash flow generation, while self-funding the de-risking plans. The early cash-flow is generated from hooking up/commissioning the tested exploration wells through trucking the produced oil from the planned Extended Well Test (EWT) project. The monetary value generated from producing these wells will fund the subsequent 3-year exploration/appraisal programs. Subsequently, the success of de-risking and commissioning of the next batch of wells will support future on-plot and off-plot facilities investments. Therefore, the strategy is designed to address challenges associated with marginal resources by coupling de-risking and development plans in a phased manner with small and areal-focused segments to allow future clustering of successful discoveries named Hub Development System. The development Phases are built on the success of preceding de-risking activities, such as exploration/appraisal wells and any dynamic information gathered from EWT projects. The number and size of development phases will rely on the success of de-risking activities, residual risks, funding appetite and potential economic return.

The Produce While Explore Strategy, which is underpinned by "Agile Approach", is becoming more common for marginal opportunities. It can be applied to both conventional and unconventional oil and

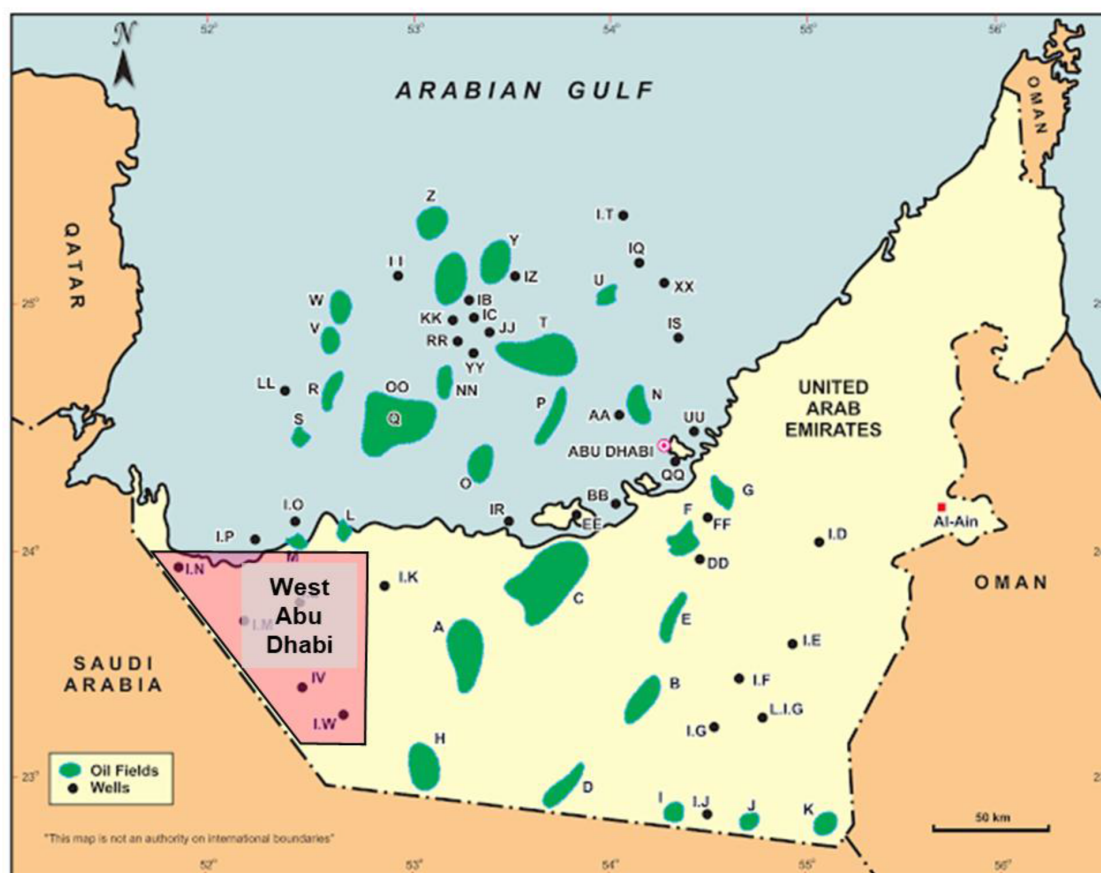
gas resources. It has the benefit of phasing the funding programs while defining the project "Exit Point" at different stages of the project depending on the outcome of the de-risking activities. Therefore, the development pace can always be controlled (accelerated, decelerated, or even paused) at any stage of the project. This approach allows establishing subsurface analogues and surface configurations that can be utilized and replicated for the subsequent phases. This strategy is proposed for the West Abu Dhabi to be implemented in the very near future.

The key to success for this strategy is having a competitive cost model on different aspects of the project; be it on drilling, facilities, or cost of technology deployment. This normally requires different contracting strategies, followed by a steep learning curve to arrive at efficient execution for the subsequent phases. The other important success factor is to have adequate technologies to enable better allocation of hydrocarbon, such as using of advanced seismic techniques and enhancing oil production through various forms of stimulations.

Instruction

Hydrocarbons exploration began in the late 19th century, and it has been gone through massive improvements as oil and gas have become the key energy component in driving the 3rd and 4th industrial revolutions. The exploration methods, which are backed by technological advancements and vast experience, have dramatically developed over the last few decades. These advancements have enabled geologists to explore hydrocarbons in challenging topographic and subsurface terrains, such as mountainous areas or deep seas, and to reach to depths close to 15 km. As exploration efforts have been progressed, most of the major and easy discoveries have been made, yet mature basins worldwide continue to deliver (Beaumont et al., 2000). However, the leftover resources are becoming challenging to explore and develop due to either the subsurface complexity or economically marginal reservoirs. Therefore, a critical strategic mindset shift is required in order to continue successfully exploring and producing hydrocarbons.

West Abu Dhabi (Fig. 1), regionally, is part of Rub' Al Khali mature basin, which contains many oil and gas discoveries/developments in Saudi Arabia, Abu Dhabi, Oman, and Qatar (Stewart, 2016). The Rub' Al Khali basin is covered with hundreds of kilometers of 3D/2D seismic data, thousands of wells, many existing analogues for different plays, numerous studies on basin modeling and regional geology. West Abu Dhabi area contains a stack of plays with either gas, condensate or oil resources that are trapped structurally in small four-way dip closures or stratigraphically in heterogeneous carbonate-evaporites deposits (Franco et al., 2018). However, some technical challenges can impose different risks that influence any proposed exploration/development strategy (Bump et al., 2021), and the small size and geological nature of the leads/prospects require a different strategic mindset, where the exploration/development strategy needs to be of a different form.



take some time and require more data, especially if no analogue is available, hence it is normally associated with a steep learning curve. Unlike the conventional four-way dip closures, where appraisal wells can provide fair definition of the opportunity by drilling 3 to 4 wells at the different flanks. Therefore, with marginal opportunities, one of the challenges to be aware of is how to strike a balance between data collection during the de-risking stage and kicking off the production phase. The Agile method, which is based on the ability/flexibility to adjust plans as projects are progressing, is thought to be more suitable to deal with exploring/developing marginal resources. Therefore, the best approach is to collect data, analyze them and move on with preset small development phases or steps. Those steps should have clearly defined exit-points in case things come way below the preset/desired values. The learner mindset is not about knowing it all upfront, but it is about building the learning as we go and learning from what is happening around us to deliver the value in reasonable timeframe.

Exploration-Production Natural Culture. Most of the major organizations face the challenge of working-in-silo, where sometimes even the team's KPIs could compete against each other. On some occasions, this competition could either compromise short-term or long-term values, while in others it could even cause value destruction. As an immediate result, resource prioritization can be a difficult task, and can lead to shortsighted decisions. The natural or organic concept calls for making the best utilization of those resources including key expertise to serve the end-to-end KPI which is linked to the long-term vision rather than focusing on some short-lived objectives. Strategy-wise, applying this concept for the Exploration/Development teams would also lead to better utilization of exploration wells for development purposes or development wells to explore and appraise other targets. Therefore, streamlining the resources to meet a bigger goal can be easily enforced and cascaded down to the individual teams through the assigned managers.

Method

The strategy adopted for West Abu Dhabi hinges on the Explore-Produce concept to facilitate quick cash-flow generation, while self-funding the de-risking plans. The early cash flow is generated from hooking-up/commissioning the tested exploration wells through trucking the produced oil from the planned Extended Well Test (EWT) project. The monetary value generated from producing these wells will fund the subsequent 3-year exploration/appraisal programs. Subsequently, the success of de-risking and commissioning of the next batch of wells will support future both on-plot and off-plot facilities investments. Therefore, the strategy is designed to address challenges associated with marginal resources by coupling de-risking and development plans in a phased manner, with small and areally focused segments, to allow future clustering of successful discoveries. This can be named the Hub Development System, and the development phases are built on the success of the preceding de-risking activities such as exploration/appraisal wells and any dynamic information gathered from EWT projects. The number and size of development phases will rely on the success of de-risking activities, residual risks, funding appetite and potential economic return.

The Produce While Explore Strategy, which is underpinned by the Agile Approach, is becoming more common for marginal opportunities. It can be applied to both conventional and unconventional oil and gas resources. It has the benefit of phasing the funding programs while defining project "Exit Point" at different stages of the project depending on the outcome of the de-risking activities. Therefore, the development pace can always be controlled, either accelerated, decelerated or even paused, at any stage of the project. This approach allows for establishing subsurface analogues and surface configurations that can be utilized and replicated for the subsequent phases. The exploration, de-risking and production activities take place at the same time, but with different magnitudes depending on the opportunities' maturation stage (Fig. 2). The facilities, commonly on lease agreement, are considered from day one, but with different scope size, where flexibility to expand, replicate, terminate or even buyout those facilities should be built-in to the contract.

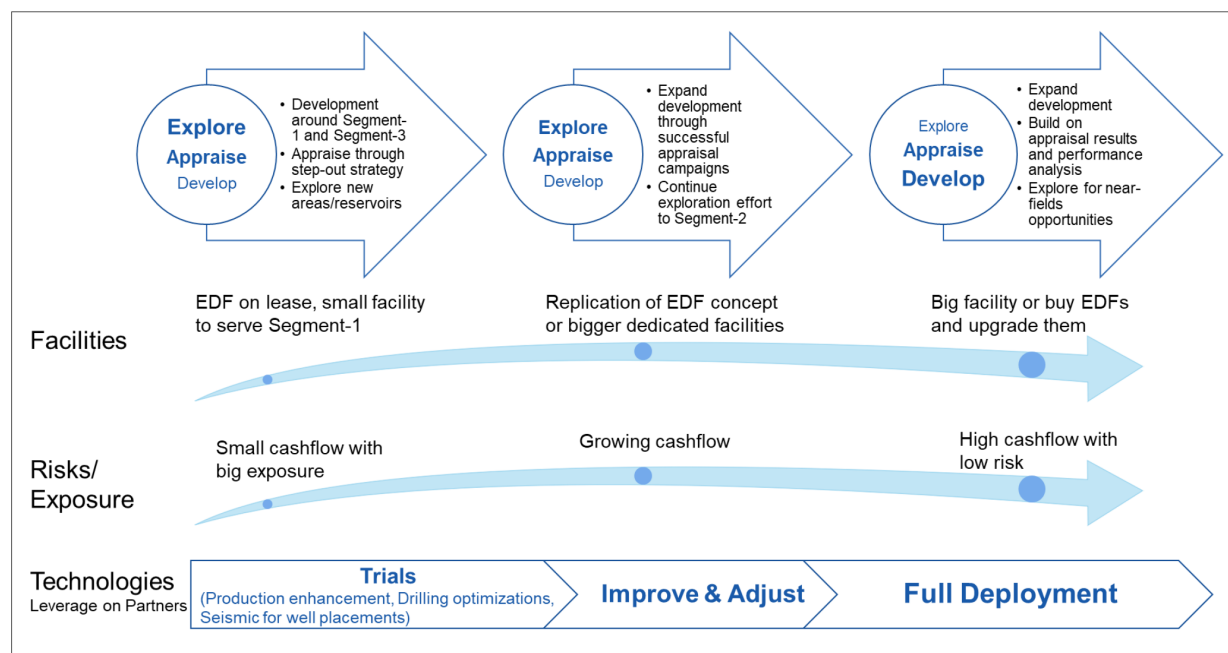


Figure 2—Concept of Explore-Appraise-Produce at different stages of the opportunity maturation

The key pillars for the proposed strategy call for the exploitation of oil and gas resources in all plays to maximize the chance of success and achieve a cost-efficient way of de-risking those resources. It couples the exploration and the development aspects that could lead to early value "production" monetization. In addition, the strategy covers the long-term value generation by addressing the key elements that make the project sustainable and economically robust, such as proposing an adequate cost model or trialing certain technologies to reduce the cost or enhance the production. The cost model needs to be competitive in different project aspects, whether on drilling, facilities, or the cost of technology deployment. This could require different contracting strategies, followed by a steep learning curve to achieve an effective execution of the subsequent phases. The other important success factor is to have adequate technologies to enable better allocation of the hydrocarbon (well location placement), such as the use of Advanced Seismic Techniques and enhancing oil production through various forms of stimulation.

The Produce While Explore Strategy for West Abu Dhabi proposes to address exploration of both oil and gas plays, while giving more attention to de-risking the oil leads/prospects. The current practice, where the total depth of most wells has just reached to the Middle Cretaceous NahrUmr formation, has been associated with clear missed opportunities. These wells could have cheaply exploited the Upper Jurassic Arab formation gas prospects, by extending the total depth to a few thousand feet, which in turn would have expedited the de-risking and the potential development of the Arab gas reservoir. Furthermore, the strategy dives into splitting the area into smaller segments that can exploit and de-risk the resource in a phased manner. The idea behind the phasing is to focus the effort on generating quick cash flow by targeting the oil prospects that require minimum funding, while in parallel, de-risking other segments. The easy path for this phasing approach is to build on the success of the existing exploration wells and expand outwards towards the least known areas. The phasing also broadly addresses the conventional and unconventional resources separately as they are associated with different technical challenges. However, where they co-exist, the de-risking strategy will address both resources together to maximize cost efficiency.

This strategy also relies on the significance of continuing the effort during the execution, where detailed activities' optimization can yield further value improvement. The optimization can cover getting the best out of the drilling sequence such as minimizing rig-move cost and horizontalization of the wells in one rig visit, well placement using seismic attributes, data-gathering, etc.

Application

The exploration strategy in West Abu Dhabi aims to target both oil and gas resources, and the hydrocarbon phase maturation of the petroleum systems can be geographically mapped broadly into three areas and divided into segments: Segment-1, Segment-2, and Segment-3 (Fig. 3). Segment-1 contains oil trapped in the Middle Cretaceous Mishrif formation carbonate and Tuwayil formation sandstone reservoirs, believed to be sourced from the Middle Cretaceous Shilaif intra-shelf basin. This area has also proven to contain gas in the Upper Jurassic Arab formation, therefore, the Mishrif/Tuwayil oil resources are considered as the primary targets, while gas resources are to be targeted as secondary. Segment-2 is interpreted to contain significant resources of unconventional oil in the thick Middle Cretaceous Shilaif formation and conventional oil resources in Tuwayil formation, which is expected to develop as turbidite channels and lobes of around 20-30 ft intervals, thinning towards west Shelf area. The Arab gas is also expected, and can be targeted as secondary objective. Segment-3 is interpreted to be predominantly associated with the Upper Jurassic Arab formation gas and the Middle Jurassic Diyab formation unconventional gas. As this area is structurally the shallowest, it is expected that Arab gas to become more wet with potential increase in the condensate volume northwards. The gas targets are considered as primary targets in this segment, while the risky Mishrif/Tuwayil oil resources are secondary targets.

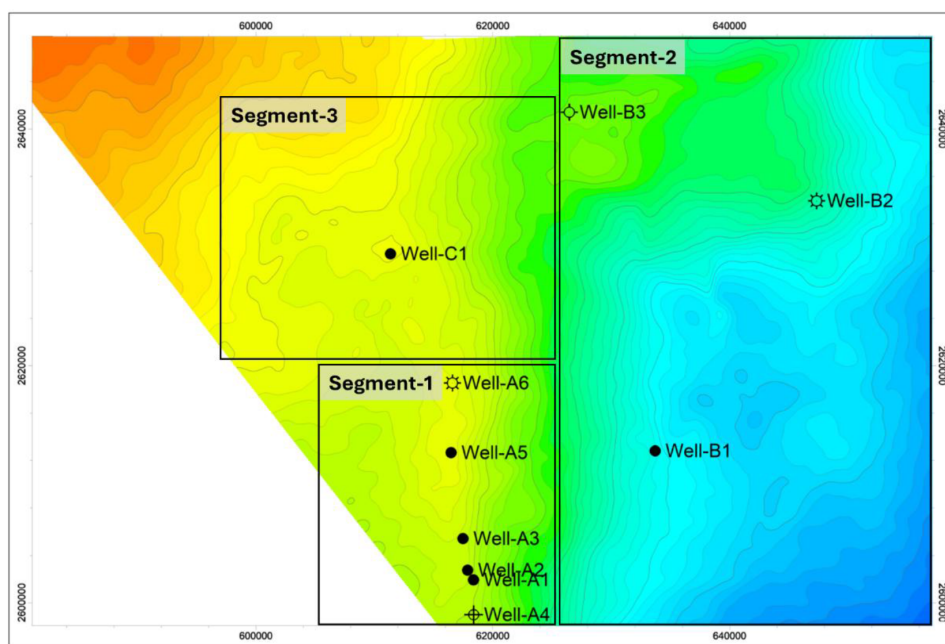


Figure 3—Three segments in West Abu Dhabi

Strategy for Segment-1

Segment-1 is considered the key area in the proposed strategy as it has the most confident data so far. This segment, which covers the Mishrif/Tuwayil prospects and leads, contains more than 70% of the conventional oil resources and has the largest number of penetrations. The three wells, Well-A1, Well-A2 and Well-A3, were drilled on the south-eastern flank of the Middle Cretaceous Mishrif carbonate shelf and preserve an oil column ranging from 50 to 75 ft above a free water level, while Well-A4 is interpreted to be below that free water level. Well-A5 was horizontalized southwards in the Middle Cretaceous Tuwayil reservoir. This segment has seen positive results from the testing of Well-A1, Well-A2, Well-A3, and Well-A5, which show a high potential. Therefore, this segment has been given the highest priority as it can yield the quickest cash flow with minimum investment.

Segment-1 De-risking Plan. In structural-trapping systems, once the free water level is established, the de-risking of the opportunities becomes straightforward, whereas in the stratigraphic-trapping cases the geological heterogeneity adds another layer of complexity (Tendil et al., 2022). Despite the lack of clarity on the definition of the trap, the wells have confirmed the hydrocarbon presence, which proves the exploration technical success for Mishrif and Tuwayil reservoirs (Xiao et al., 2022). Therefore, the key risks, including the reservoir quality and connectivity, are mainly associated with commerciality. The reservoir quality is clearly demonstrated in the variability of the well tests, and the heterogeneous nature of the reservoir poses a threat to connectivity which could affect the oil recovery. However, the key exploration risks for the prospects and leads include two aspects. The first risk can be identified by trap definition and lateral sealing, as these are mostly stratigraphic traps that remain the risk of hydrocarbon presence (Jewell et al., 2019). The second risk can be named as access to charging, because some leads/prospects are located in migration-path shadow.

The de-risking strategy for Segment-1 is prepared to address both the exploration and development risks. The strategy builds on the high confidence obtained from the well tests at Prospective Area-1 (Fig. 4), where only one appraisal well is proposed to examine the extent of the accumulation to the west. To prove the commerciality of Prospective Area-1 discoveries, the EWT of Well-A1, Well-A2, and Well-A3 is highly recommended as it will provide performance data on the production sustainability of the Mishrif and Tuwayil reservoirs. These dynamic data, then, can be used to assess the reservoir connectivity and the pool size. It is proposed to horizontalize Well-A1 and Well-A2 after the EWT. Due to the heterogeneous nature of the reservoir, horizontal wells are believed to be more efficient in connecting the different rock types and increasing the well productivity. The lateral holes will also be utilized for the proposed development plan. The accessibility and integrity of the wells need to be checked ahead of the EWT and the horizontalization, and accordingly, a workover plan should be put in place.

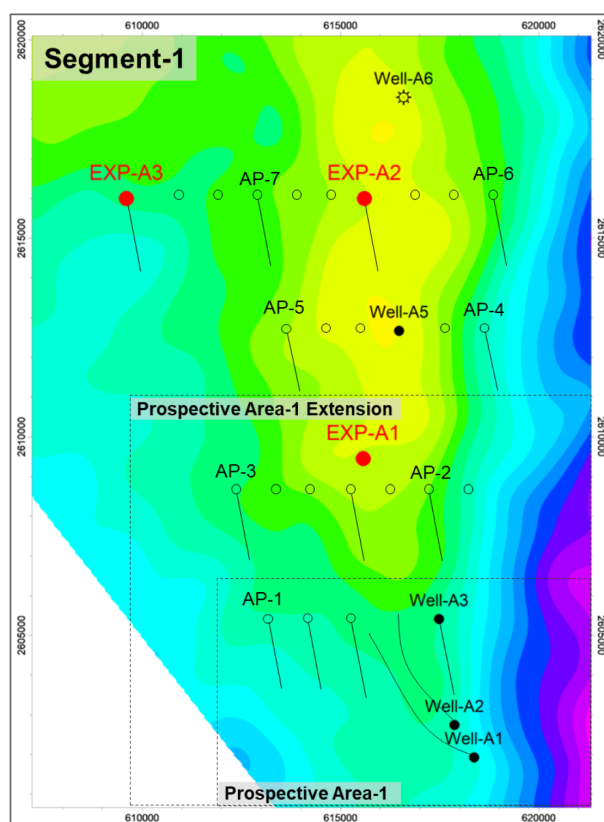


Figure 4—De-risking plan for Segment-1 showing the location of exploration and appraisal wells

For the higher-risk prospects/leads to the north of Prospective Area-1, three exploration wells are proposed to be drilled. Strategically, ranking of the three wells is done to facilitate development expansion from South to North. The first well, EXP-A1, is placed between Well-A3 and Well A5. Upon the success case of this well, two appraisal/development wells are proposed on the eastern and western sides. These two wells would delineate the so-called "Prospective Area-1 Extension" as they will provide valuable information on the extent of the oil accumulation to the west, and the reservoir quality within the area. That information can also be calibrated with geophysical data to provide more confidence in predicting the reservoir presence to place the future development wells. The north of Segment-1 has already been discovered by Well-A5, which targeted Mishrif and Tuwayil and was horizontalized in Tuwayil. Therefore, this area requires two other appraisal wells, on the sides of Well-A5, to expand the development northwards.

The other two exploration wells are proposed just to the south of Well-A6. The exploration well, EXP-A2, will be placed on pre-existing four-way-dip closure that can target the Upper Jurassic Arab formation and Lower Cretaceous Thamama groups, as well as the stratigraphic Mishrif and Tuwayil traps up-dip of Well-A6. It is worth noting that Well-A6 was drilled on 2D seismic data and located on the flank of this structure according to the available 3D seismic. The Arab gas discovery was proven in Well-A6, and some oil shows were observed in Mishrif and Thamama intervals. This proposed exploration well will also be associated with two appraisal wells AP-6 and AP-7 which would provide a deeper understanding of the limit of the hydrocarbon accumulations and reservoir distribution in this segment, hence facilitating expanding the development to cover the entire segment. The other exploration well, EXP-A3, is proposed on the northwestern side of Segment-1 which has been ranked low due to the high charge risk, hence it is recommended to consider drilling the appraisal wells AP-6 and AP-7 before drilling this well. It is worth remembering that the strategy calls for deepening those exploration wells and having pilot holes in some of the appraisal wells to provide cost-effective means of exploiting the potential of the deeper opportunities, such as Arab, Diyab and the Permian Khuff formations.

Segment-1 Notional Development Plan. Regarding these types of heterogeneous and low-permeability carbonate rocks, horizontal wells are considered to maximize oil production through connecting the different rock types. Notionally, until the FDP study is conducted, the proposed development for Mishrif/Tuwayil will be 2000-3000 m long horizontal wells, spaced at around 1000 m apart. Depending on the degree of reservoir connectivity, infill drilling to 500 m spacing can always be considered to increase the recovery factor. However, the project economics should be run to support the feasibility of the additional CAPEX. The infill wells provide the flexibility to produce them as depletion wells or convert them to water injection wells to support the line-drive waterflood. Placement of the horizontal wells will be at the top of either Mishrif or Tuwayil reservoirs, depending on which one has better reservoir development. Seismic data and attributes could play a major role in better placement of these horizontal wells and minimize sidetracking of the development wells. It must be stated that at this stage, communication between these two units will not be understood until depletion profile is established during the development.

The notional development of Segment-1 has been designed to build on the previously discussed de-risking plan (Fig. 5). For the 1000m-spacing case, the development plan could consist of 38 wells, where only 24 new development wells will be needed. The other 14 wells would either exist or be horizontalized as per the success of the de-risking plan. In the absence of sustainable test or solid performance data, it is rather challenging to establish a robust commercial development case. Therefore, several well-profile cases are estimated to generate production profiles that can feed into the economic analysis. The well profiles, with initial rates of 150, 500, 800, 1000 BPD, are declined aggressively so that the wells would only last for 10 years or so to mimic poor reservoir connectivity scenario (Fig. 6). The massive variation in peak production, ranging from 4.5 to 30 MBPD, signifies the need for the EWT data that can reduce the uncertainty in those profiles. The expected resources to be recovered range from 11 to 60 MMBO depending on the estimated well productivity.

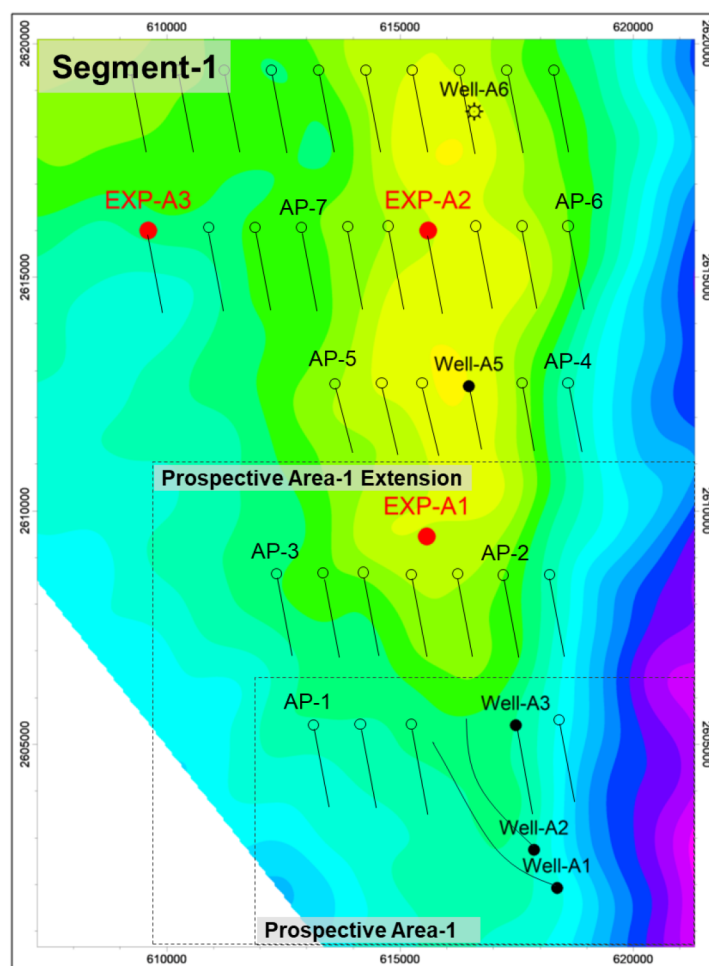


Figure 5—The notional development plan for Segment-1

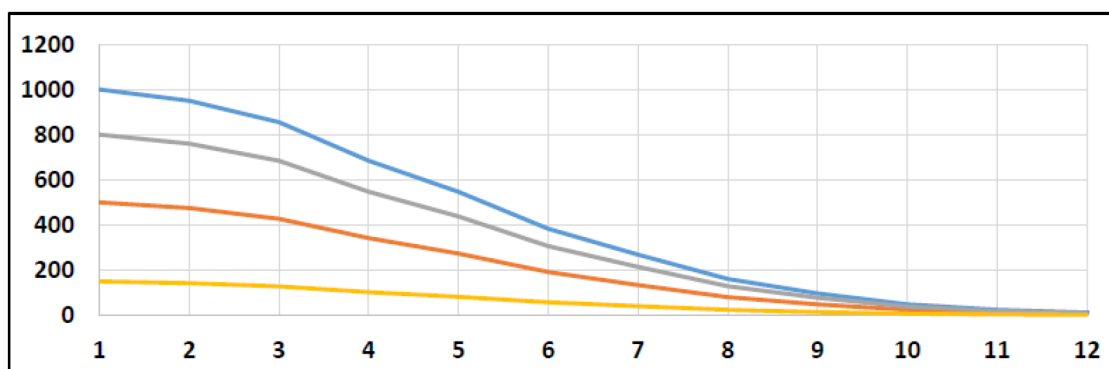


Figure 6—The well profile cases used to estimate the production profiles for Segment-1

Despite the uncertainties in the production profiles, the agile approach that this strategy hinges on, can be used to support the Explore-Produce concept in Segment-1. Instead of pursuing the development of the full segment, it can be divided into a series of FDPs, starting from Prospective Area-1 at the far South, then expanding to Prospective Area-1 Extension in the center, before reaching the North of Segment-1. This approach minimizes investment and provides Exit Points at each stage of the de-risking/development. These options are considered as part of this strategy (Table 1), and the EWT options are the most affordable and can provide a quick return.

Table 1—Considered Options for developing the resources in Segment-1.

Options		Benefits and Reward	Expected Investment
Option-1	EWT of Tested Vertical Wells	Cheap and Quick Oil(2 to 3 MBPD)	Cost of Trucking
Option-2	EWT of Horizontalized Wells	More Sustainable Rate(2 to 3 MBPD)	Cost of Trucking and 2 horizontalizations
Option-3	Development (Depletion) of Prospective Area-1	8 MMBO of Oil(6 MBPD)	Cost of Small facility (6 MBPD), 120K Pipeline and 2 Extra Wells
Option-4	Development (Depletion) of Prospective Area-1 Extension	16 MMBO of Oil(9 MBPD)	Cost of Bigger facility (9 MBPD), 120K Pipeline and 9 Extra Wells
Option-5	Development (Depletion) of full Segment-1	55 MMBO of Oil(25 MBPD)	Cost of Bigger facility (25 MBPD), 120K Pipeline and 34 Extra Wells

As this project would rely on trucking, the CAPEX investment would be very minimal, while the main expenditure would be the OPEX which can reach \$35 million. The contract can be designed in such a way that it can cease at any point, hence the real exposure is only 10 \$ million. Option-2 calls for horizontalization of Well-A1 and Well-A2 to provide more sustainable production, which is still a cheap project considering the overall low cost of the project. However, this option can be considered if Option-1 becomes economically unattractive due to unsustainable production rates.

Option-3, -4 and -5 are relatively more expensive options, hence the exposure would be slightly higher. For Option-3, which covers the development of Prospective Area-1, one appraisal well and two development wells are notionally proposed for the depletion case. For the water-flood case, three oil producers and two water injectors are proposed. In both cases, as the drilling CAPEX can be phased out for 2 to 3 years, the real exposure is the investment in the facilities, and oil and gas pipelines. The facility would be on a lease agreement; therefore, the actual exposure pertains solely to the cost of the pipelines. However, if the EWT (Option-1) goes ahead, the decisions on Option-3 investment can be deferred until robust data is established. Therefore, the Exit Point would be the cost of the EWT. In addition, performance data from the EWT would also reduce the uncertainty range of the profiles plotted and would allow better design of the facility contracting strategy. For this option, the expected facility capacity will be around 5 to 6 MBPD.

Option-4 proposes to expand the development northwards towards Well-A5 which is called Prospective Area-1 Extension. The exploration well, EXP-A1 and horizontalization of Well-A1 and Well-A2 are required to facilitate the development expansion. Upon the successful discovery of the proposed well, strategically, the development can step out away from the well towards the East and the West following the 1000-m spacing plan, or drill two appraisal wells at the edge, and then infill inwards towards the exploration well. For this option, 9 additional wells are proposed to develop the block, and the production profiles show massive range depending on the assumed well rate. For this option, the expected CAPEX exposure for the Exit Point would be mainly the cost of the exploration, and the investment for the pipelines can be limited to the design stage only, while the construction can be timed with the analysis of the EWT outcomes.

Option-5 builds on the success of Options-3 and -4. Therefore, the proposed 38 horizontal wells to cover this segment assume that the de-risking will be conducted as per the suggested plan. The final shape and size of this development could vary depending on the outcomes of the preceding activities, and the exposure associated with the Exit Point could be the cost of the 3 exploration wells and the 7 appraisal wells. However, the agile approach provides flexibility in accelerating or decelerating development of the individual smaller phases depending on the pace of the de-risking plan and approved expenditure. In the case of slow de-risking pace, the Capex exposure can be limited and controlled, but the NPV will also be reduced.

For the water-injection case, additional 36 water injectors would be required in a potentially line-drive pattern (Fig. 7). However, to test the concept of the waterflood, a pilot around Well-A2 may be needed to establish the response at the proposed 500-spacing. Drilling the remaining water injectors will depend

on the performance of the oil producers and depletion profile, hence more likely they will follow similar phasing plan.

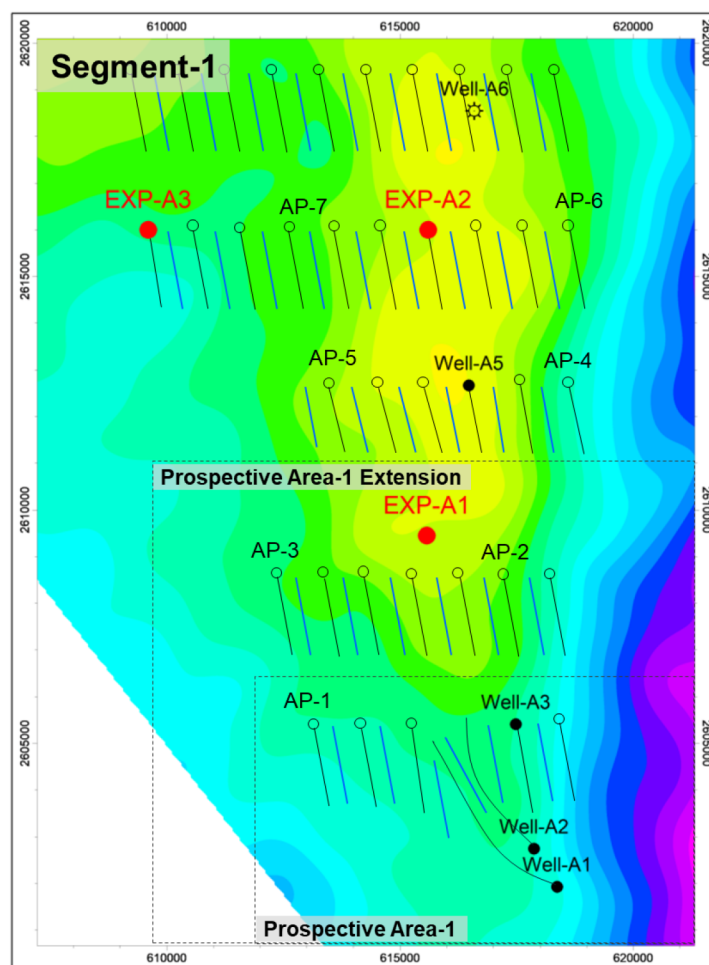


Figure 7—Proposed notional Waterflood for Segment-1

Strategy of Segment-2

Segment-2, namely Prospective Area-2, covers mainly unconventional Shilaif and conventional Tuwayil leads and prospects, and contains huge volumes of oil resources. One exploration well, Well-B1, located on the western edge of the basin, has shown some promising potential in the lowest bench.

Segment-2 De-risking Plan. For the unconventional shale development, productivity is the major risk. As per the Eagle Ford Shale analogue, productivity can be assessed through drilling a 10,000-foot horizontal well with 50 fracture stages, pumping 100 tons of proppant. Nevertheless, as an immediate de-risking step, it is proposed to work over the vertical well Well-B1, and proceed with the fracture program and test the well. The following step will be to horizontalize the wells and perform the fracture as per the study proposal to enable good assessment for the FDP.

The Tuwayil opportunity in Segment-2 is interpreted to be dominated by turbidite and fan-lobe deposits of around 20ft thick. The key risk associated with Tuwayil prospects is trap definition and lateral sealing, as these are very settled structures, hence the hydrocarbons can continue migrating westward. The de-risking plan for Tuwayil proposes drilling one well, EXP-B1, at the southeastern part of the basin, which can also examine Shilaif and other opportunities such as Mauddud and Arab. Depending on the results of the wells and the strategic direction, this well can be either kept for the Tuwayil or the unconventional Shilaif developments. For the latter option, horizontalization and the fracturing job will need to be planned as per

the proposed unconventional strategy. The Follow-up exploration wells (EXP-B2, EXP-B3 and EXP-B4) will be proposed depending on the results of the workover of Well-B1 and EXP-B1 (Fig. 8).

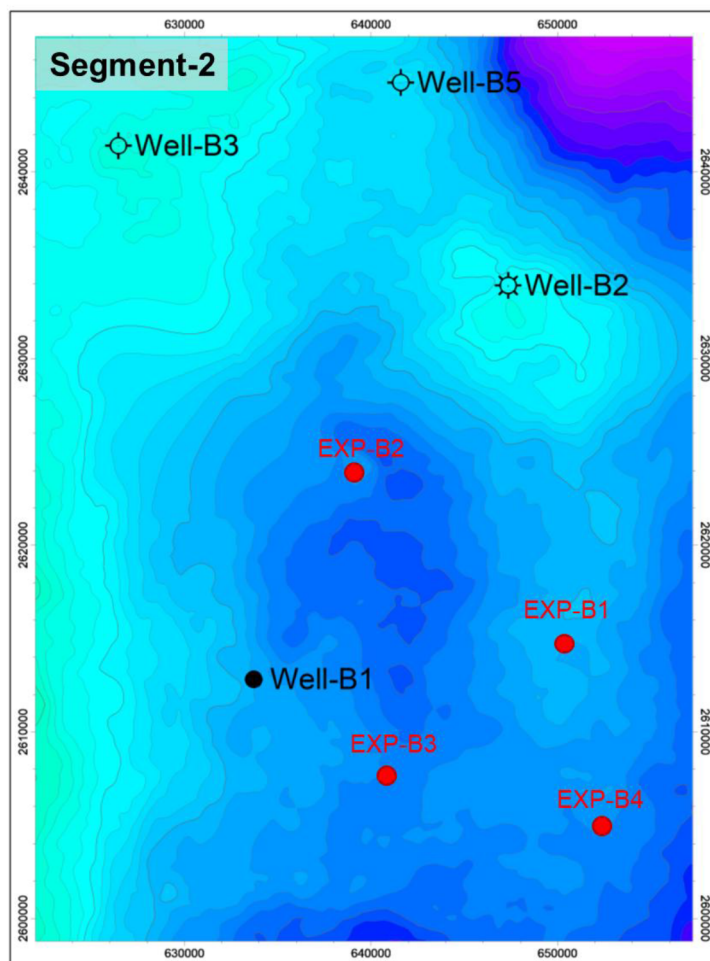


Figure 8—De-risking plan for Segment-2 showing the location of exploration and appraisal wells

Segment-2 Notional Development Plan. The Shilaif formation consists of three benches, and the lowest bench is expected to be more mature and to have most of the recoverable resources. If those resources are targeted in one single phase, the estimated peak production rate will be more than 100 MBPD. This production is expected from 904 10,000-foot-long horizontal wells spaced at 610 feet (less than 200m), and with 50 fracture stages pumping 100 tons of proppant per stage.

Considering the challenging nature of this development, and the anticipated massive facility and well CAPEX, the development can strategically be designed to minimize the big upfront investment and improve the learning curve as the development progresses. This can be achieved by building on the success of the fracture job of Well-B1 and horizontalizing the well. This activity will initiate at least 9 development wells around Well-B1 with total production of 1000 BPD. The next stage will be to capitalize on the exploration wells, which will enable drilling another 27 wells, and the wells will cover the four directions of the basin (Fig. 9). Therefore, from this point onwards, the pace of the subsequent development will depend on the outcome of those 36 development wells, and the size of the investment. In-filling the wells to cover the full basin (900 wells) will require a few years and many rigs and strong technical team. Therefore, the drilling contracting strategy will need to be in place to facilitate the big phase of drilling. It is essential to bear in mind that some of these wells might not be successful, either due to subsurface challenges or operational issues. This clearly means the strategy relies on achieving the best out of the mass number of wells (namely,

the production from most of the wells will compensate for the poorer wells), hence early panic and reaction could kill the project so early in the journey.

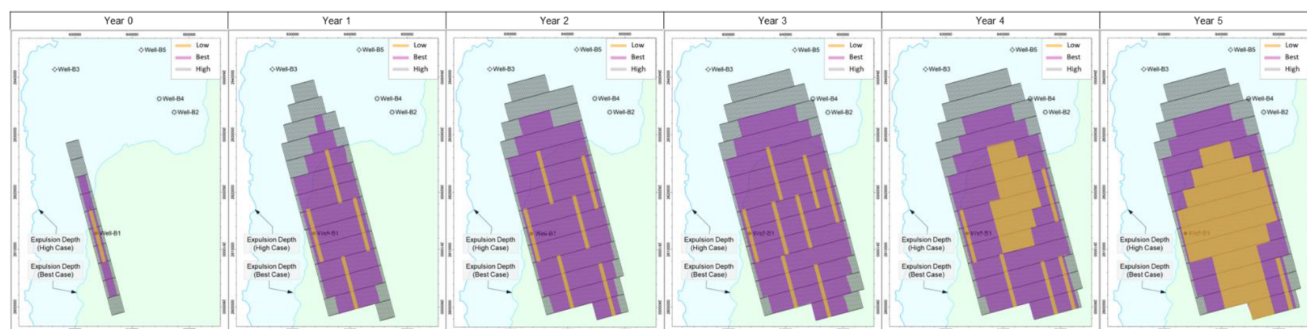


Figure 9—Proposed notional development plan for Segment-2

On the facility-front, the first two stages of the project will not require major investment; however, the limited production profile of a few thousand barrels might not support this investment. Therefore, for this stage of development to economically succeed, it will need to be combined with conventional development. The production from the first two stages of unconventional developments can be easily absorbed by the second and third conventional FDPs. However, for this to be fully synchronized, the following enablers are needed:

- De-risking activities of Segment-1 and Segment-2 to be conducted as per the plan to facilitate the FDPs,
- Forming an experienced, multidisciplinary taskforce from 2026 to detail the options discussed in this strategy, and analyze the outcomes of the de-risking activities to proceed with subsequent activities,
- Learning from ongoing activities, especially on unconventional technologies and drilling practices, and
- Seeking means to lower the cost such as revisiting the drilling contracting, especially for unconventional resources.

Strategy of Segment-3

Segment-3 is given a lower priority in the proposed strategy as it has limited data with only one exploration well, Well-C1. The de-risking and development of this segment can be deferred to later years to establish some technical and commercial learnings from Segment-1. Like Segment-1, this segment contains Mishrif/Tuwayil prospects and is expected to contain less than 30% of the conventional oil resources.

Segment-3 De-risking Plan. Segment-3 contains the northern part of the carbonate Shelf, and is expected to have both Mishrif and Tuwayil reservoirs developed in this segment with same discussed reservoir quality and connectivity risks. Well-C1 has confirmed the presence of hydrocarbon in this area, which provides strong technical confidence in the reliability of the petroleum system along the shelf from Prospective Area-1 at the South up to Well-C1 at the North. The access-to-charge and lateral sealing risks will still be applicable for areas north of west of Well-C1.

The de-risking strategy for Segment-3 relies on sidetracking Well-C1 to address the reservoir quality away from the wellbore, and hence the well productivity. In addition, an exploration well is proposed to the south of Well-C1 to prove the hydrocarbon presence on the ridge between Well-A6 and Well-C1. The well will target Mishrif/Tuwayil and Thamama reservoirs as oil and Arab intervals as secondary gas targets. Upon the success case of this well, five appraisal/development wells are proposed on the eastern and western sides of this segment, which are expected to facilitate the development of the entire segment (Fig. 10).

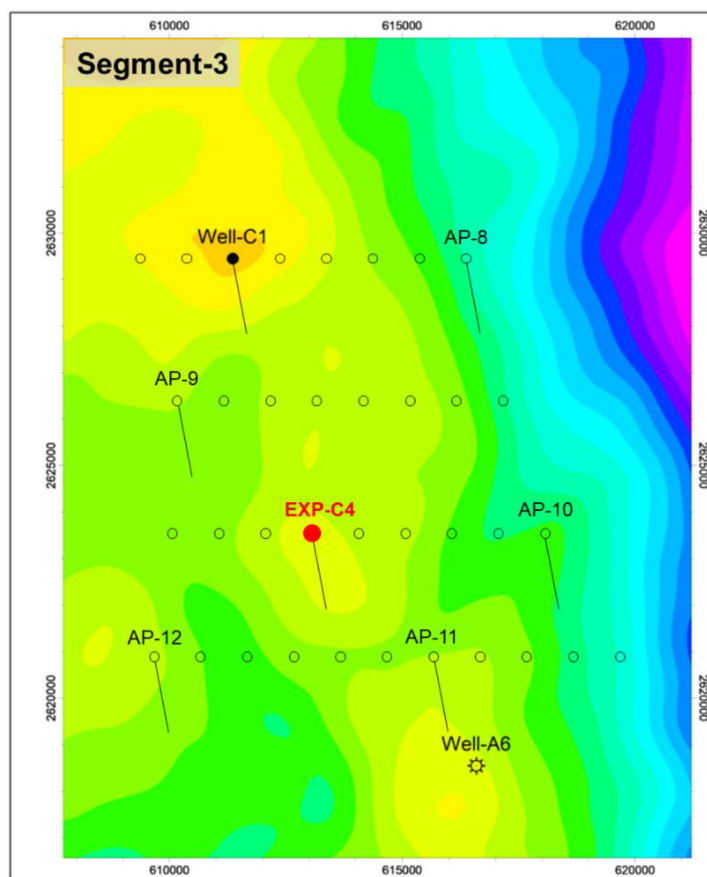


Figure 10—De-risking plan for Segment-3 showing exploration and appraisal wells

Segment-3 Notional Development Plan. The national development for Mishrif/Tuwayil reservoirs will be 2000-3000 m long horizontal wells spaced around 1000 m apart. This depletion development can be followed by 500 m spacing water-flood development to increase the recovery factor. Early indications suggest that Segment-3 could not be as prolific as Segment-1. Therefore, the well-profile cases are limited to 100, 300 and 500 BPD, and are declined aggressively so that the wells would only last for 10 years or so to mimic a poor reservoir connectivity scenario profiles. The massive variation in peak production, ranging from 2.5 to 14 MBPD, signifies the need for the EWT data that can reduce the uncertainty in those profiles. The expected resources to be recovered range from 7 to 34 MMBO depending on the estimated well productivity.

The notional development of Segment-3 builds on the de-risking plan. For the 1000m spacing case, the development plan could consist of 36 wells, where 29 new development wells will be needed, and the other 7 wells would have been drilled through the de-risking plan. For the water-injection case, an additional 40 water injectors would be required in a potentially line-drive pattern. However, the implementation of the water flood project will take place after establishing communication between the producers. In addition, it will be updated based on the success of Segment-1 waterflood project.

Opportunities and Risks

As this strategy intends to consolidate the basis for maximizing the value from West Abu Dhabi, there are other opportunities that are yet to be discovered. The Middle Cretaceous Maaddud formation has been proven in other areas in Abu Dhabi whose reservoir is characterized by laterally distributed tight reservoirs. However, the reservoir quality in West Abu Dhabi is not as good as other areas, so the potential can be assessed by testing productivity with and without Frac in any future opportunity, and can be utilized as a

fallback option for any dry well. The Lower Cretaceous Thamama Intervals have been explored in this area for many years, but no discovery has been made yet. It is realized that the source rock might be a key risk for this play, however, the opportunity can always be assessed in the upcoming activities, especially well EXP-A2, which is expected to penetrate the structure closure at all these intervals. There are other opportunities in this area worth mentioning such as Khuff and Pre-Khuff plays which are expected to be filled with gas.

There are two key risks associated with the exploration and development of the oil resources in West Abu Dhabi. The access to charge towards the Northern and the Northwestern part in West Abu Dhabi can be mitigated by focusing on the eastern side, and then progressively moving towards the risky areas as better understanding is established. In addition, aligning gas and oil exploration strategies would increase the exploration chance of success at competitive cost. The second risk is the low well production rate which would affect the commerciality of the resources. At this stage, the rates from Prospective Area-1 are very promising, but the limited test duration has been triggering some concerns around the sustainability of these rates. This risk can be mitigated by performing the needed EWT to provide better estimation of oil production rates. In addition, horizontal wells with production enhancement technologies can always be considered to improve productivity and commerciality.

Notional Economics

The Produce while Explore Strategy releases the importance of cost competitiveness and optimization in increasing the value of the resources. The cost optimization model focuses on three aspects: strategic saving, activities optimization and well optimization.

The strategic saving addresses the high-value items that are normally covered at the project or FDP level. The proposed ideas would include:

- Aligning oil and gas exploration/development strategy to maximize value generation,
- Considering surface facilities to be sourced on a Rental Basis with minimum initial investment,
- Making use of nearby operators for sharing facilities and liquid evacuation means,
- Utilizing cheaper horizontal wells to explore/appraise complex plays and heterogeneous reservoirs,
- Using cheap rigs or coil tubing to drill top-hole sections while having the main rigs drill the dual laterals,
- Drilling slim wells to target deeper exploration intervals, and
- Considering campaign drilling in a grid-pattern following the Streets concept of Urban Plan, with dual-lateral wells drilled from the same pad targeting two targets.

The activities optimization, such as drilling sequence optimization, is the second aspect of cost competitiveness, and it is normally addressed during the execution stage through the business planning cycle. The optimization will cover:

- Drilling sequences to facilitate campaign drilling and campaign well testing,
- Aligning drilling activities with other nearby operators to reduce mobilization cost, and
- Horizontalizing wells from the first rig visit after the vertical hole to avoid double mobilization and hole-entry cost.

The well optimization would consider:

- Optimizing data gathering and linking it with the Value of Information (VoI),
- Maximizing the use of the seismic attribute analysis to minimize drilling multiple holes chasing the reservoir during well placement, and

- Assessing the possibility of moving rigs with the Mast up during the development phase to minimize rig-move cost and rig-up and rig-down durations.

Economic analyses are performed for different purposes at different stages of the opportunity maturation process. During the exploration stage, the economic analysis is considered to assess the drilling worthiness of the prospect, and to support the de-risking activities. At this stage, the uncertainties are at the greatest level, hence many assumptions are used to perform the task. The oil in place and the well rates are the greatest uncertainties. In addition, the cost estimates are associated with limited degrees of accuracy, and they will become more reliable as the project scope gets clearer. Nevertheless, economic attempts are documented in detail in in-house study report, and others are reported here to shed some lights on the commerciality of the conventional and unconventional opportunities in West Abu Dhabi. The cost optimization will be done for all projects, prior to any major decision such as tendering once the scope is fully defined.

Segment-1 EWT of Well-A1, Well-A2 and Well-A3 Economics. The EWT has been proposed to cover the key wells in Prospective Area-1: Well-A1, Well-A2 and Well-A3. It aims to massively de-risk the subsequent developments, while providing some cash flow to self-fund some of the exploration activities. The limited scope of the EWT, which is intended to limit the HSE exposure, is expected to result in marginal return. There are two cases that are considered to prove some insights about this initiative. The 3-year production case shows a positive NPV, while continuing with EWT for the 5-year production will yield more than tens of millions NPV. The 5-year case is run to support extending the tests until the first FDP facilities and pipelines are ready. It is essential to realize that these cases are prepared for feasibility purposes, hence the cost, especially the CAPEX, are expected to drop down after the optimization.

Segment-1 Prospective Area-1 Notional Economics. The economic exercise for Prospective Area-1 development has also been attempted. Until the EWT is performed, the uncertainties associated with the production profiles are not expected to be narrowed. For feasibility purposes, economic cases for an average well rate of 500BPD and 1000BPD are performed. The following assumptions are used to generate those cases:

- Facility on rent, hence CAPEX is mainly for pipelines and road construction,
- 6 wells in total (one existing, two to be horizontalized and 3 new wells),
- Depletion only,
- Artificial Lift (ESP) is included for the 6 wells, and
- Same cost profiles used for both cases.

For the above assumptions, it becomes apparent that the 500BPD for Prospective Area-1 is not feasible, while the 1000BPD is very robust. Obviously, with further optimization, the 500BPD case may economically pass, but will always remain marginal. The 800BPD has not been run, but it is expected to also be economically robust. The economic results of Prospective Area-1 waterflood cases are extremely sensitive to the well profile and number of wells. Three oil producers and two water injectors, spaced at 1000m, have been found to form optimal waterflood development.

Segment-2 Unconventional Shilaif Notional Economics. The unconventional Shilaif notional economic analysis assumes that the full development will be executed in one phase over few years, and all infrastructure and facility cost will be covered by this development. This development is expected to be very CAPEX intensive as it will contain up to 900 wells. The economic sensitivities show that development becomes feasible if the well cost is dropped down to less than \$12 million for a production case of around 100MBPD. The unconventional development is expected to be split into few phases, and more likely will be combined with other ongoing developments. In addition, as the learning curve is established, a massive

reduction in the cost of the facilities and the well is expected to materialize before going for the main development phase.

Conclusions

In conclusion the Produce While Explore Strategy for West Abu Dhabi addresses significant prospective resources (around a billion barrels of oil and many TCFs of gas), that can yield a value of a few billion dollars, but it is associated with some genuine challenges:

- Stratigraphic plays with limited technical experience, especially on unconventional development,
- Slow exploration/appraisal pace covering such vast area, on the contrary, these activities can be done more aggressively to decide on the way forward, and
- It can be a CAPEX intensive development in view of the nature of geology and expected development.

The strategy is designed to adopt a different mindset that is suited for marginal opportunities based on Explore-Produce concept. It aims to focus on generating early cash flow from Segment-1 and building a learning curve for the unconventional Shilaif oil in Segment-2. Segment-1 by itself can be divided into smaller exploration-development areas such as Prospective Area-1 and Prospective Area-1 Extension. These smaller areas would allow minimizing the investment, while maximizing the geological and commercial understanding of the area. The learning from this segment can then be implemented in Segment-3, which is now considered slightly more challenging.

The success of the Strategy hinges on the following recommendations:

- Coupling oil and gas exploration to provide cost-efficiency de-risking and maximize the chance of success.
- Considering vertical wells for exploring the stacked plays, followed by horizontal legs to prove commerciality through maximizing production rate with production enhancement technologies if required.
- Performing EWT to mitigate production-sustainability risk and provide key data on reservoir connectivity and for designing the facilities.
- Assessing Early Development Facility (on Lease Strategy) option for Prospective Area-1 Development and replicating for other phases.
- Liaising with partners to maximize the technical knowledge and strive for more competitive facilities, pipelines, and road construction cost.
- Driving well cost down through adopting different contracting strategies and campaign drilling to make the project more economically robust and enable efficient execution.

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